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A. P. SHAH INSTITUTE OF TECHNOLOGY
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(Religious Jain Minority)

Department of Computer Science & Engineering (AI & ML)

"Empowering Smart Living: A Secure and Sustainable Voice-Driven Home Automation System"

Group members with Student Id

Sarang Bahikar (22106129)
Krishit Doshi (22106001)
Gaurav Kolambe (22106085)
Mohammed Ali Bardi (22106058)

Project Guide
Prof. Monali Korde

Outline

- Introduction
- Literature Survey of the existing systems
- Limitations of the existing systems
- Problem statement and Objective
- Proposed System
- Experimental Setup(Details about input/ dataset, Performance evaluation parameters, Software and hardware setup)
- Implementation(Screenshots of all modules)
- Conclusion and future work
- References
- Research Paper Publication / Patent/ Copyright status

Introduction

- The system focuses on creating smarter living spaces by integrating voice-enabled controls with advanced automation technologies.

- This system enhances convenience and accessibility through natural language interaction, while prioritizing robust security to safeguard user data and device access.

- With a strong emphasis on sustainability, it enables energy-efficient management of lighting, appliances, climate, and security systems, reducing both energy consumption and costs.

- By combining AI, IoT, and voice assistants, the system delivers a user-friendly, future-ready solution that transforms homes into secure, sustainable, and intelligent environments.

Literature Survey of the existing system

Author Name	Title	Technologies used	Findings
Prashanth Kura, et al Year: 2025 Publication: IEEE	Application of Deep Learning Techniques for Internet of Things (IoT) Device-Enabled Smart Home Automation with Predictive Modeling.	Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Long Short-Term Memory Networks (LSTMs), Predictive Modeling	The study reveals that applying deep learning techniques in IoT-enabled smart home automation significantly enhances accuracy, efficiency, and intelligent decision-making. By integrating IoT device data, it provides real-time insights and adaptive behavior, ensuring improved personalization, comfort, and security for users.

Author Name	Title	Technologies used	Findings
W. Sarada, et al Year: 2025 Publication: IEEE	IoT in Smart Homes: Enhancing Lifestyle Through Intelligent Automation and Control	Home Assistance, IFTTT (If This Then That), Amazon AWS IoT	The research highlights that IoT integration in smart homes greatly improves lifestyle by enabling seamless automation, remote control, and real-time monitoring of household devices. It enhances convenience, safety, and energy efficiency while offering personalized experiences tailored to user needs.

Author Name	Title	Technologies used	Findings
P. Anusha, et al Year: 2024 Publication: Taylor & Francis	Unveiling the potential: An in-depth analysis of Empowering smart homes through IoT-based sensing and monitoring platforms for enhanced automation	Microcontrollers /SOCs: ESP8266, ESP32, Zigbee / Z-Wave, Smart Switches & Plugs: Enable on/off control	The research finds that IoT-based sensing and monitoring platforms play a crucial role in enhancing home automation by providing real-time data collection, analysis, and intelligent control of devices. The study also highlights that IoT platforms enable seamless connectivity between devices.

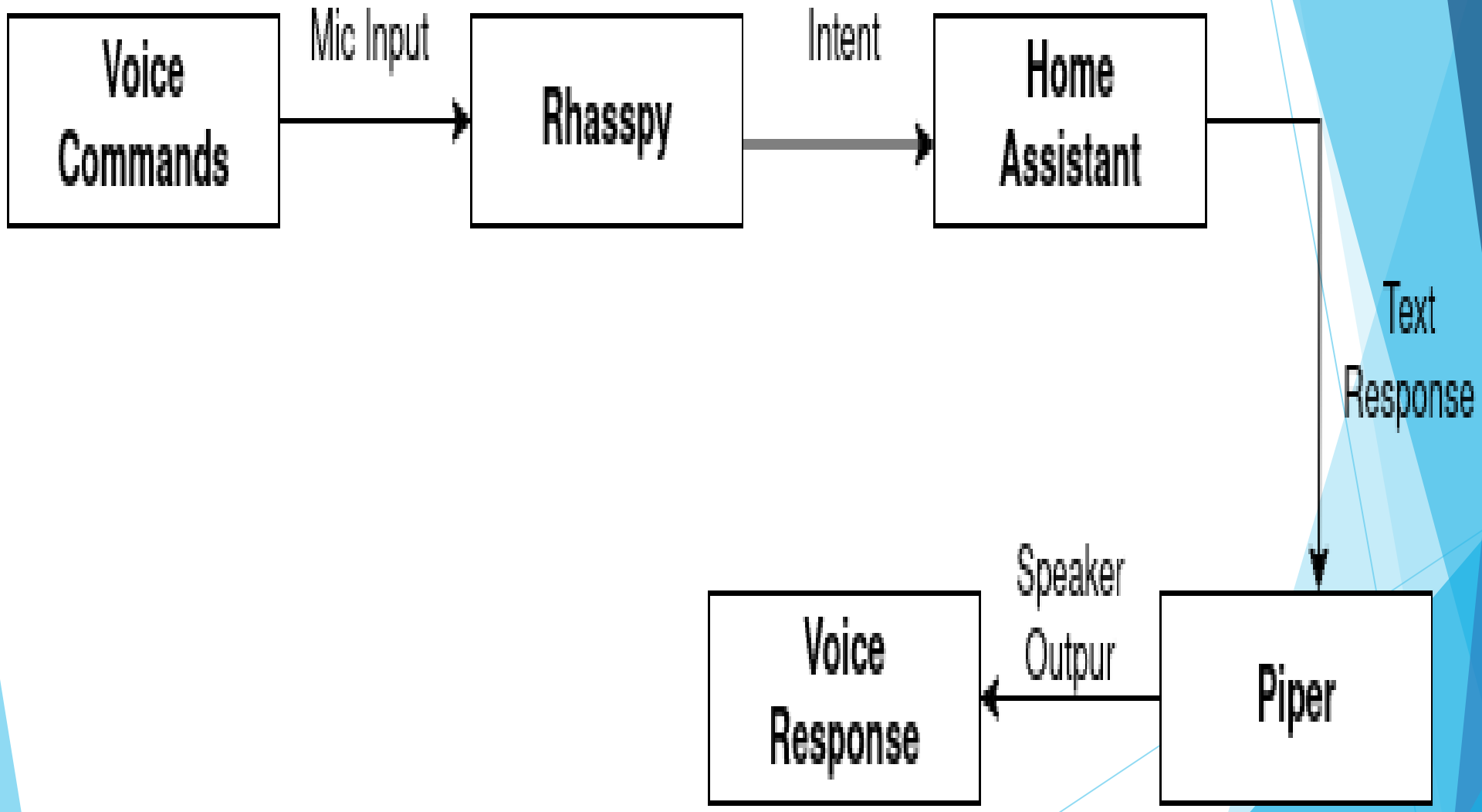
Limitations of Existing Systems

- High Initial Cost – Installation and setup may be expensive.
- Dependence on Internet – Requires stable connectivity for seamless operation.
- Privacy Concerns – Risk of data breaches and unauthorized access.
- Limited Compatibility – Not all devices or appliances may integrate smoothly.
- Voice Recognition Challenges – Issues with accents, background noise, or misinterpretation.

Problem statement

- The solution runs a Raspberry Pi, ensuring modularity, easy deployment, and efficient resource usage. The system prioritizes local processing, security, and extensibility, supporting various integrations and automations while keeping the environment self-contained. This approach delivers an intelligent, voice-driven interface for controlling home devices without reliance on external cloud services, ensuring privacy, reliability, and flexibility for end users.

System Design



Technologies and methodologies

Technologies:

- Raspberry Pi 4
- Docker
- Docker Compose
- Home Assistant

Methodologies:

1. Modular Architecture:

   Each component runs in its own container for separation of concerns

2. Offline-first Processing:

   All voice recognition and automation logic are handled locally without cloud dependencies

3. Scalability and Extensibility:

   New devices, commands, and services can be integrated by extending the configurations

Implementation

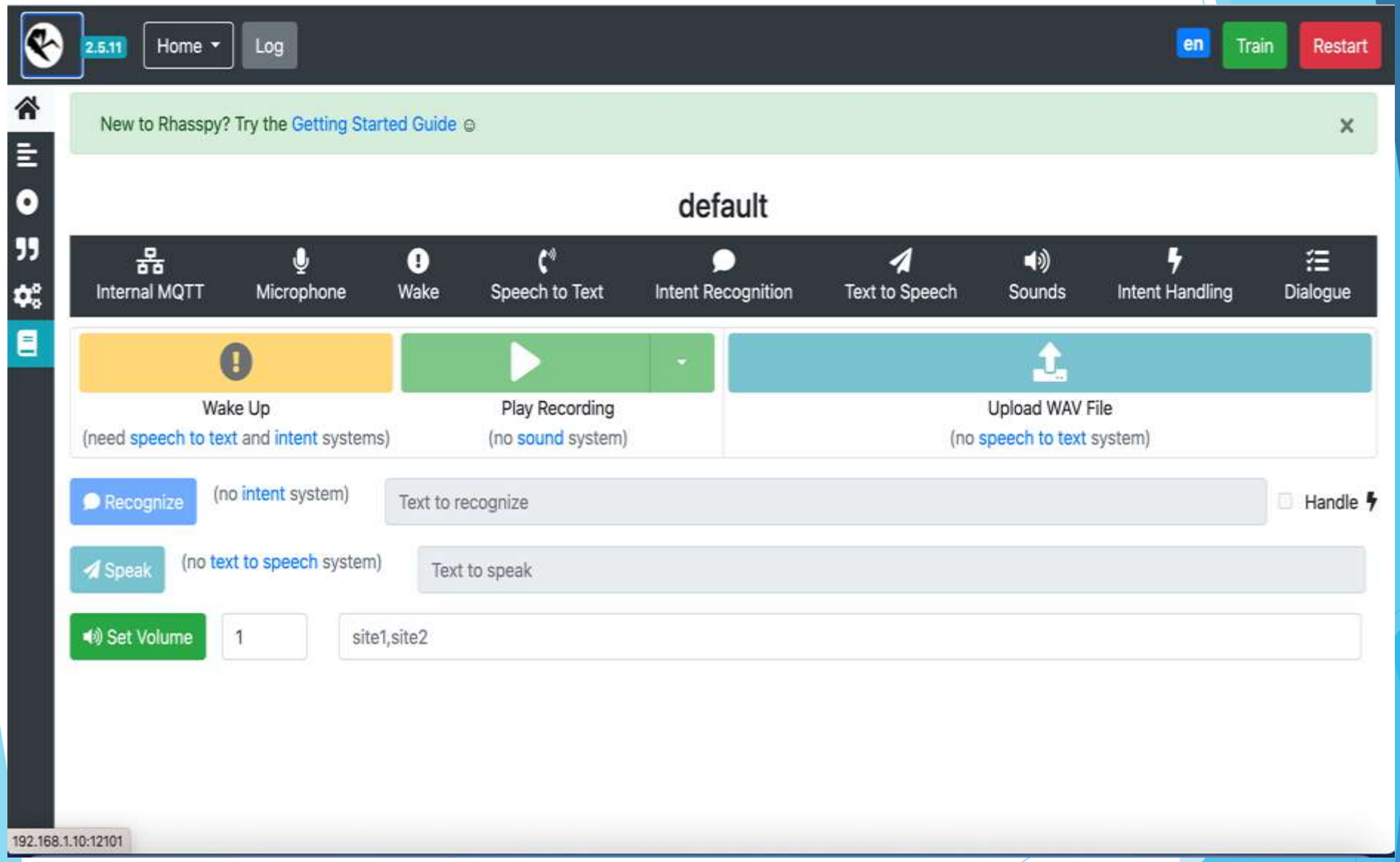


Figure 1: Rhasspy Dashboard

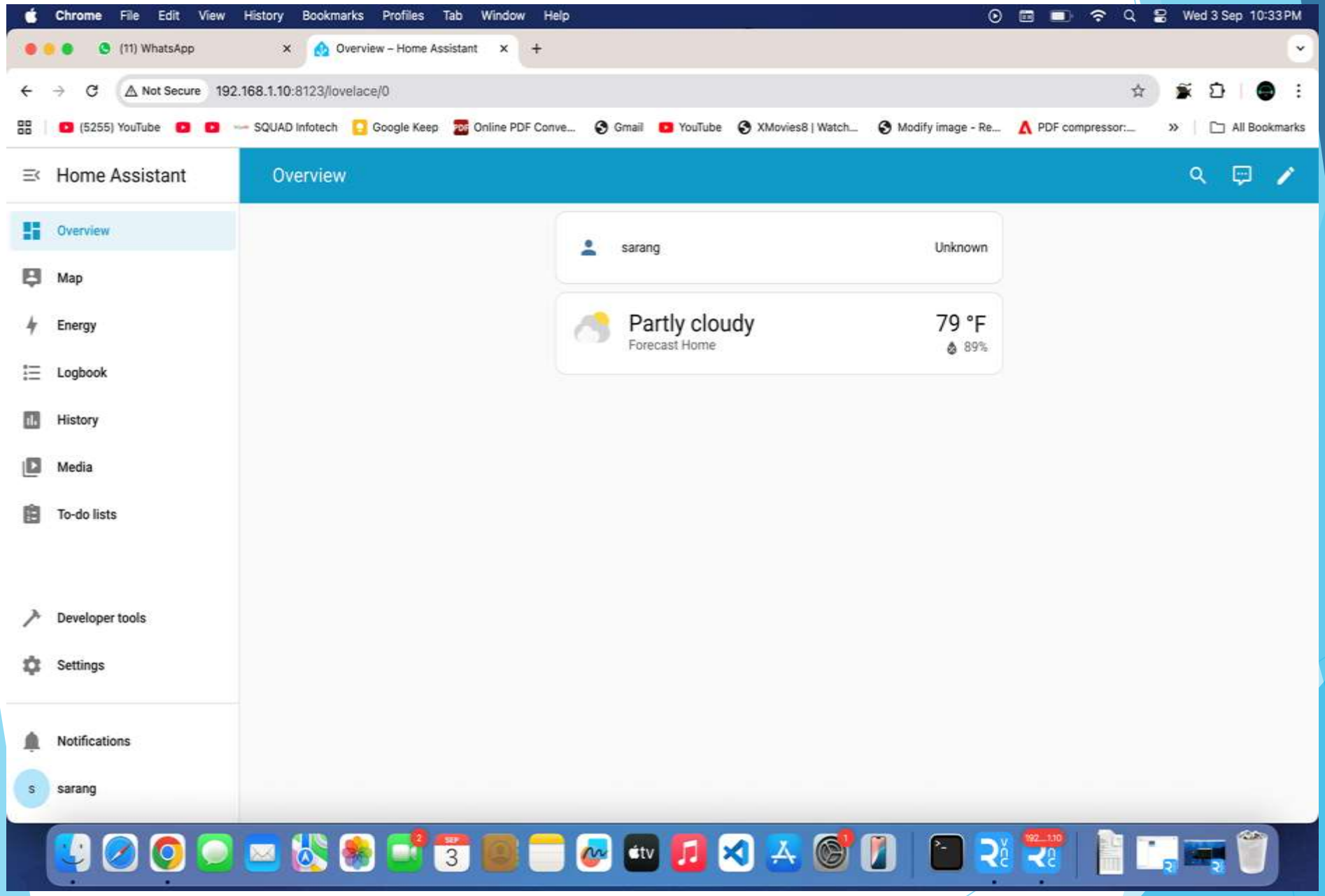


Figure 2: Home Assistant

Conclusion

- This demonstrates how smart technology can transform modern households into intelligent, energy-efficient, and secure living spaces. By integrating voice-enabled controls with AI and IoT, the system enhances convenience, accessibility, and user experience while ensuring robust data security.

References

[1] **Prashanth Kura, Rajashekhar Chitimalla (2025).** Application of Deep Learning Techniques for Internet of Things (IoT) Device-Enabled Smart Home Automation with Predictive Modeling.

Application of Deep Learning Techniques for Internet of Things

[2] **W. Sarada, K.V. Prashanth, A. Beneta Mary, K. Premkumar, Vinodh P Vijayan, Radha G (2025).** IoT in Smart Homes: Enhancing Lifestyle Through Intelligent Automation and Control.

IoT in Smart Homes: Enhancing Lifestyle Through Intelligent Automation and Control.

[3] **P. Anusha, P. Durga Vijay, V. Reethika, G.G.S. Karthikeya, Vijay Kumar Burugari (2024).** Unveiling the potential: An in-depth analysis of Empowering smart homes through IoT-based sensing and monitoring platforms for enhanced automation.

Unveiling the potential: An in-depth analysis of Empowering smart homes

Thank You...!!